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The Guardian – Hate Speech Detection Module Evaluation & Validation Report

This document summarizes the training, validation, and evaluation results of the Hate Speech Detection Module developed as part of the research project **"The Guardian: A Modular AI for Proactive Human-Layer Cybersecurity on Social Media."** The module uses a fine-tuned DistilBERT Transformer model trained on a curated dataset of labeled social media posts.

Dataset Distribution

Dataset Split	Number of Samples
Training Set	19,826
Validation Set	2,478
Test Set	2,479
Total Samples	24,783

Model Training Details

Model: DistilBERT (distilbert-base-uncased)
Training Method: Transformer fine-tuning using supervised learning.
Epochs: 3
Task: Multiclass classification for detecting hate speech and toxic language in social media text
Frameworks: Python, HuggingFace Transformers, PyTorch

Validation Results

Metric	Validation Result
Loss	0.2784
Accuracy	0.9169 (91.69%)
Precision (Macro)	0.7854
Recall (Macro)	0.7835
F1 Score (Macro)	0.7842
Evaluation Runtime	422.12 seconds
Samples per Second	5.87

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Final Test Evaluation Results

Metric	Test Result
Loss	0.2592
Accuracy	0.9201 (92.01%)
Precision (Macro)	0.7834
Recall (Macro)	0.7845
F1 Score (Macro)	0.7831
Evaluation Runtime	416.03 seconds
Samples per Second	5.96

Conclusion

The trained model achieved a **test accuracy of 92.01%** and a **macro F1 score of 0.783**, demonstrating strong performance in detecting hate speech while maintaining balanced precision and recall across classes. Evaluation was conducted using a held-out test dataset that was not used during training to ensure unbiased performance measurement.